Yijie Ma

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Skills

C/C++, Go, Python, Distributed Systems, AI Infrastructure (GenAI, MCP, A2A, GPU/RDMA), gRPC, API Gateway/L7 Proxy, Performance Optimization, OpenTelemetry, Storage & Filesystems (RocksDB), Kubernetes, Networking

Work Experience

Google April 2017 - Present

Software Engineer

Google Cloud API Gateway & Service Infrastructure

- Led technical direction and C++ distributed tracing architecture for Gemini Agent Platform across HTTP/gRPC, defining OTel SemConv data models for Cloud, GenAI, Model Context Protocol (MCP), and Agent-to-Agent (A2A).
- Drove cross-org alignment across Auth, Observability, and API Gateway teams to resolve ambiguous agentic requirements; scoped milestones and mentored junior engineers on multi-quarter observability projects.
- Built end-to-end W3C Trace Context propagation, trace span generation, trace context initiation and sampling, and asynchronous OTLP trace/log export pipelines for a C++ L7 proxy handling millions of QPS.
- Scoped and led multi-quarter delivery of an isolated fast API serving release track across control-plane and serving teams, decoupling rapid AI and Cloud feature rollouts from the slower stable release track.
- Pioneered LLM multi-agent workflows using reusable AI Skills to automate release risk evaluation, qualification reporting, and staging promotions, adopted across the team.

gRPC Open Source Remote Procedure Call Framework

- Core developer and Maintainer of gRPC (40K+ GitHub stars) with 150+ authored PRs, driving open-source RFCs and PR reviews across C++ core, networking, and observability.
- Co-authored gRFC A79 to standardize non-per-call telemetry; architected the C++ templatized Metrics framework and OpenTelemetry plugin, cutting Counter overhead by 10%.
- Designed and built the cross-platform EventEngine asynchronous DNS resolver across Linux/POSIX and Windows.
- Designed POSIX fork syscall handling and resolved multi-threaded pre-fork deadlocks across gRPC C++ and Python.

Google Kubernetes Engine (GKE) Networking

- Engineered GKE control-plane automation in Go to auto-provision multi-NIC RDMA network topologies (VPCs, subnets, IP ranges) for AI-optimized GPU clusters.
- Designed concurrent request handling with distributed transactions and validated multi-node GPU communication using NVIDIA's NCCL test suite.

Google Drive Desktop Application

- Designed and built the rsync-style differential upload and Gzip stream compression encoders with compressibility heuristics to save upload bandwidth for millions of users.
- Architected the USB device backup subsystem backing up millions of files daily, increasing backup speed by 10x (140s to 18s for 1GiB) by optimizing I/O read buffers.
- Added Crashpad crash reporting on Linux, built a post-mortem crash debugging tool, and resolved a new category of Windows crashes (1000+ daily) in threads running on heap-allocated stacks.
- Designed the cross-platform Preferences system supporting user-level settings and enterprise admin policy overrides across macOS and Windows.

Stadia Cloud Gaming Service

- Increased game click-to-play launch speed by 75% (1.58s to 396ms) via CPU profiling and container setup optimization.
- Designed a Linux Cgroup library for isolating and configuring container resources on cloud gaming VMs.

NetApp August 2015 - March 2017

Member of Technical Staff

Worked on the AltaVault Cloud-Integrated Storage device. Designed and built a filesystem software to backup user's files to multi-cloud vendors. Designed and implemented full and incremental snapshot backup and restore functions using RocksDB.

Education

Carnegie Mellon University Sep 2013 - May 2015

Master of Science in Electrical and Computer Engineering

Coursework: Introduction to Computer System, Embedded Realtime System, Advanced Storage System, Internet Service, Cloud Computing, Distributed System

Donghua University Sep 2009 - May 2013

Bachelor of Engineering in Electrical Engineering and Automation

Awards: 2012 IEEE-standard Micromouse competition national award issued by China Computer Federation (CCF)